Lecture 01 – Optimization Algorithm

Stat 5376: Statistical Machine Learning

Ruiqi Liu

Department of Mathematics and Statistics
Texas Tech University

http://www.math.ttu.edu/~ruiqliu/

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Review of Linear Algebra

- Let $x = (x_1, \ldots, x_p)^\top \in \mathbb{R}^{p\times 1}$ and $A = (A_{ij}) \in \mathbb{R}^{p\times p}$.
- Linear Form:

$$Ax = \begin{pmatrix} \sum_{j=1}^p A_{1j}x_j \\ \vdots \\ \sum_{j=1}^p A_{pj}x_j \end{pmatrix}.$$

- Quadratic Form:

$$x^\top Ax = \sum_{i=1}^p \sum_{j=1}^p x_i x_j A_{ij}.$$

- Spectral Radius: For $A \in \mathbb{R}^{p\times p}$ and let $\lambda_1, \ldots, \lambda_p$ be the eigenvalues, then we define

$$\rho(A) = \max\{|\lambda_1|, \ldots, |\lambda_p|\}.$$

Derivative

- For a mapping $f : \mathbb{R}^k \to \mathbb{R}^p$, we denote

$$f(x) = \begin{pmatrix} f_1(x) \\ \vdots \\ f_p(x) \end{pmatrix} \text{ for } x \in \mathbb{R}^k.$$

- The derivative of f is

$$\dot{f}(x) = \frac{df(x)}{dx} = \begin{pmatrix} \frac{\partial f_1(x)}{\partial x_1} & \cdots & \frac{\partial f_1(x)}{\partial x_k} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_p(x)}{\partial x_1} & \cdots & \frac{\partial f_p(x)}{\partial x_k} \end{pmatrix} \in \mathbb{R}^{p \times k}.$$

- When $p = 1$, the derivative is a row:

$$\dot{f}(x) = \begin{pmatrix} \frac{\partial f(x)}{\partial x_1} & \cdots & \frac{\partial f(x)}{\partial x_k} \end{pmatrix} \in \mathbb{R}^{1 \times k}.$$

- Gradient: $\nabla f(x) = \dot{f}^\top(x) \in \mathbb{R}^{k \times 1}$ is a column.

Second Derivative

- When $p = 1$, the derivative is a row:

$$\dot{f}(x) = \begin{pmatrix} \frac{\partial f(x)}{\partial x_1} & \dots & \frac{\partial f(x)}{\partial x_k} \end{pmatrix} \in \mathbb{R}^{1 \times k}.$$

- When $p = 1$, we can take second derivative

$$\ddot{f}(x) = \frac{d\dot{f}^\top(x)}{dx} = \begin{pmatrix} \frac{\partial^2 f(x)}{\partial x_1^2} & \dots & \frac{\partial^2 f(x)}{\partial x_1 \partial x_k} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f(x)}{\partial x_k \partial x_1} & \dots & \frac{\partial^2 f(x)}{\partial x_k^2} \end{pmatrix} \in \mathbb{R}^{k \times k}.$$

- Hessian matrix: $\nabla^2 f(x) = \ddot{f}(x)$.

Taylor Expansion

- Recall: if $f : \mathbb{R} \to \mathbb{R}$, we have

$$f(x) = f(a) + f^{(1)}(a)(x-a) + \frac{1}{2!}f^{(2)}(a)(x-a)^2 + \frac{1}{3!}f^{(3)}(a)(x-a)^3 + \ldots.$$

- If $f : \mathbb{R}^d \to \mathbb{R}$, we have

$$f(x) = f(a) + (x-a)^\top \nabla f(a) + \frac{1}{2}(x-a)^\top \nabla^2 f(a)(x-a) + \ldots.$$

- For higher order term, we need tensor expression.

Chain Rule

- Why derivative is a row?
- $f : \mathbb{R}^k \to \mathbb{R}^p$ and $h : \mathbb{R}^p \to \mathbb{R}$.
- Composition: $g(x) = h(f(x)) = h(f_1(x), \ldots, f_p(x)) : \mathbb{R}^k \to \mathbb{R}$.
- Verify $\frac{\partial g}{\partial x_1} = \sum_{i=1}^p \frac{\partial h}{\partial f_i} \frac{\partial f_i}{\partial x_1}$
- Similarly $\frac{\partial g}{\partial x_j} = \sum_{i=1}^p \frac{\partial h}{\partial f_i} \frac{\partial f_i}{\partial x_j}$ for $j = 1, \ldots, k$.
- The derivative is $\dot{g} = \begin{pmatrix} \sum_{i=1}^p \frac{\partial h}{\partial f_i} \frac{\partial f_i}{\partial x_1} & \cdots & \sum_{i=1}^p \frac{\partial h}{\partial f_i} \frac{\partial f_i}{\partial x_k} \end{pmatrix} \in \mathbb{R}^{1 \times k}$.
- Notice that $\dot{h} = \begin{pmatrix} \frac{\partial h}{\partial f_1} & \cdots & \frac{\partial h}{\partial f_p} \end{pmatrix} \in \mathbb{R}^{1 \times p}$.
- And $\dot{f} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_k} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_p}{\partial x_1} & \cdots & \frac{\partial f_p}{\partial x_k} \end{pmatrix} \in \mathbb{R}^{p \times k}$.
- Chain rule: $\dot{g} = \dot{h}\dot{f} \in \mathbb{R}^{1 \times k}$.

Matrix Derivative

- Let $x = (x_1, \ldots, x_p)^\top \in \mathbb{R}^{p\times 1}$ and $A = (A_{ij}) \in \mathbb{R}^{q\times p}$.
- Consider $f(x) = Ax : \mathbb{R}^p \to \mathbb{R}^q$.
- By linear algebra, it follows that

$$f(x) = \begin{pmatrix} f_1(x) \\ \vdots \\ f_q(x) \end{pmatrix} = \begin{pmatrix} \sum_{j=1}^p A_{1j}x_j \\ \vdots \\ \sum_{j=1}^p A_{qj}x_j \end{pmatrix}.$$

- $\frac{\partial f_1(x)}{\partial x_1} = A_{11}$, $\frac{\partial f_1(x)}{\partial x_2} = A_{12}$, $\ldots$, $\frac{\partial f_1(x)}{\partial x_p} = A_{1p}$.
- Similarly, $\frac{\partial f_k(x)}{\partial x_1} = A_{k1}$, $\frac{\partial f_k(x)}{\partial x_2} = A_{k2}$, $\ldots$, $\frac{\partial f_k(x)}{\partial x_p} = A_{kp}$ for $k = 1, \ldots, p$.
- Derivative is

$$\dot{f}(x) = \begin{pmatrix} \frac{\partial f_1(x)}{\partial x_1} & \cdots & \frac{\partial f_1(x)}{\partial x_p} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_q(x)}{\partial x_1} & \cdots & \frac{\partial f_q(x)}{\partial x_p} \end{pmatrix} = \begin{pmatrix} A_{11} & \cdots & A_{1p} \\ \vdots & \ddots & \vdots \\ A_{q1} & \cdots & A_{qp} \end{pmatrix} = A \in \mathbb{R}^{q\times p}.$$

Matrix Derivative

- Let $x = (x_1, \ldots, x_p)^\top \in \mathbb{R}^{p\times 1}$ and $A = (A_{ij}) \in \mathbb{R}^{p\times p}$.
- Consider $f(x) = x^\top A x = \sum_{i=1}^p \sum_{j=1}^p A_{ij} x_i x_j : \mathbb{R}^p \to \mathbb{R}$.
- $f(x) = A_{11}x_1^2 + \sum_{i=2}^p A_{i1} x_i x_1 + \sum_{j=2}^p A_{1j} x_1 x_j + \text{ something}.$
- $\frac{\partial f(x)}{x_1} = 2A_{11}x_1 + \sum_{i=2}^p A_{i1}x_i + \sum_{j=2}^p A_{1j}x_j = \sum_{i=1}^p A_{i1}x_i + \sum_{j=1}^p A_{1j}x_j.$
- Similarly, for $k = 1, \ldots, p$, we have
$$\frac{\partial f(x)}{x_k} = \sum_{i=1}^p A_{ik}x_i + \sum_{j=1}^p A_{kj}x_j.$$
- Therefore, the derivative is

$$\begin{aligned}
\dot{f}(x) &= (\sum_{i=1}^p A_{i1}x_i + \sum_{j=1}^p A_{1j}x_j, \ldots, \sum_{i=1}^p A_{ip}x_i + \sum_{j=1}^p A_{pj}x_j) \\
&= (\sum_{i=1}^p A_{i1}x_i, \ldots, \sum_{i=1}^p A_{ip}x_i) + (\sum_{j=1}^p A_{1j}x_j, \ldots, \sum_{j=1}^p A_{pj}x_j) \\
&= (A^\top x)^\top + (Ax)^\top = x^\top (A + A^\top) \in \mathbb{R}^{1\times p}.
\end{aligned}$$

- If $A^\top = A$ is symmetric, then $\dot{f}(x) = 2x^\top A$.

Matrix Derivative

- Consider $f(x) = (c - Ax)^\top B(c - Ax)$.
- Let $h(x) = c - Ax$ and $g(h) = h^\top Bh$, so $f = g(h)$
- $\dot{h} = -A$ and $\dot{g} = h^\top(B^\top + B)$.
- So $\dot{f} = \dot{g}\dot{h} = -h^\top(B^\top + B)A = -(c - Ax)^\top(B^\top + B)A$.
- If $B = I$, $f(x) = (c - Ax)^\top(c - Ax)$, we can see
$$\dot{f}(x) = -2(c - Ax)^\top A.$$
- So the gradient is $\nabla f(x) = -2A^\top(c - Ax)$.
- Calculate $\nabla f(x) = 0$ leads to $A^\top Ax = A^\top c$.
- Solution is $x^* = (A^\top A)^{-1}A^\top c$.
- Linear regression: $\sum_{i=1}^n (y_i - x_i^\top \beta)^2 = (Y - X\beta)^\top(Y - X\beta)$, where

$$Y = \begin{pmatrix} y_1 \\ \vdots \\ y_n \end{pmatrix} \in \mathbb{R}^{n\times 1} \text{ and } X = \begin{pmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{pmatrix} \in \mathbb{R}^{n\times p}.$$

Linear System

- In many applications, we need to solve the following linear system of equations

$$Ax^* = b, \text{ where } A \in \mathbb{R}^{p\times p} \text{ and } x^*, b \in \mathbb{R}^{p\times 1}.$$

- Here $x^* = A^{-1}b$ is the solution.
- Linear Regression: $f(\beta) = (Y - X\beta)^\top (Y - X\beta)$
- Gradient $\nabla f(\beta) = -2X^\top (Y - X\beta)$.
- The minimizer satisfies $X^\top X\beta^* = X^\top Y$.
- So the solution $\beta^* = (X^\top X)^{-1} X^\top Y$.

Iterative Algorithm

- A^{-1} is difficult to obtain.
- Consider decomposition $A = M - N$, with M^{-1} is easy to calculate.
- So $b = Ax = (M - N)x = Mx - Nx$ implies

$$x = M^{-1}Nx + M^{-1}b = M^{-1}(M - A)x + M^{-1}b = x + M^{-1}(b - Ax).$$

- Consider iteration: $x^{(k+1)} = x^{(k)} + M^{-1}(b - Ax^{(k)})$.
- Convergence if $\rho(M^{-1}N) < 1$.
- Intuition: M, N, A, b are all scalars, then

$$x^{(k+1)} = M^{-1}Nx^{(k)} + M^{-1}b \text{ is like geometric series.}$$

- Geometric series converge if $|r| = |M^{-1}N| < 1$.

Jacobi's Method

- $A = D - L - U$.
- D is a diagonal matrix.
- L is a lower triangular matrix.
- U is a upper triangular matrix.
- Example:

$$A = \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}, \quad D = \begin{pmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{pmatrix}$$

- and

$$L = \begin{pmatrix} 0 & 0 & 0 \\ -a_{21} & 0 & 0 \\ -a_{31} & -a_{32} & 0 \end{pmatrix}, \quad U = \begin{pmatrix} 0 & -a_{12} & -a_{13} \\ 0 & 0 & -a_{23} \\ 0 & 0 & 0 \end{pmatrix}$$

Jacobi's Method

- $M = D$ and $N = L + U$.
- Advantage: M^{-1} is easy to calculate.
- Disadvantage: difficult to guarantee $\rho(M^{-1}N) < 1$.
- A Sufficient Condition for Convergence:

$$|a_{ii}| > \sum_{j \neq i} |a_{ij}|, \text{ for all i .}$$

Relaxation Method

- For $w \in (0, 2)$, let $M = \frac{D}{w} - L$ and $N = \frac{1-w}{w} D + U$
- Notice M is a lower triangular matrix, its inverse can be easily calculated.
- Example: $\begin{pmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{pmatrix} \begin{pmatrix} \widetilde{a} & 0 & 0 \\ \widetilde{b} & \widetilde{c} & 0 \\ \widetilde{d} & \widetilde{e} & \widetilde{f} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$
- $a\widetilde{a} = 1 \Rightarrow \widetilde{a} = \frac{1}{a}$.
- $c\widetilde{c} = 1 \Rightarrow \widetilde{c} = \frac{1}{c}$.
- $b\widetilde{a} + c\widetilde{b} = 0 \Rightarrow \widetilde{b} = -\frac{b\widetilde{a}}{c} = -\frac{b}{ac}$.
- $f\widetilde{f} = 1 \Rightarrow \widetilde{f} = \frac{1}{f}$.
- $e\widetilde{c} + f\widetilde{e} = 0 \Rightarrow \widetilde{e} = -\frac{e\widetilde{c}}{f} = -\frac{e}{cf}$.
- $d\widetilde{a} + e\widetilde{b} + f\widetilde{d} = 0 \Rightarrow \widetilde{d} = -\frac{d\widetilde{a}+e\widetilde{b}}{f} = -\frac{cd-be}{acf}$.

Relaxation Method

- Advantage: Easy to check $\rho(M^{-1}N) < 1$.
- A Sufficient Condition for Convergence: If A is symmetric, positive definite, and $w \in (0, 2)$, then the relaxation method converges.
- In many applications, A satisfies the above condition.
- Regression Problem: $X^\top X\beta^* = X^\top Y$ or $\frac{1}{n}X^\top X\beta^* = \frac{1}{n}X^\top Y$
- In random design, $x_1, \ldots, x_n$ are i.i.d. sampled from x.
- Notice

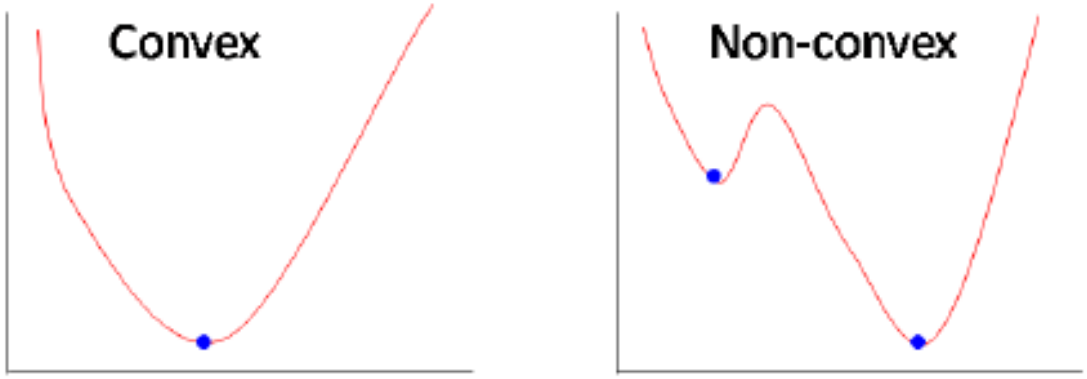
$$X = \begin{pmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{pmatrix}, X^\top = \begin{pmatrix} x_1 & \ldots & x_n \end{pmatrix}, A = \frac{1}{n}X^\top X = \frac{1}{n}\sum_{i=1}^{n} x_i x_i^\top.$$

- By S.L.L.N, we can see $A \to E(xx^\top)$ almost surely as $n \to \infty$.
- If the eigenvalues of $E(xx^\top)$ are all greater than zero, then A is almost surely positive definite when sample size n is large.

Convexity

- A function $f(x)$ is called convex if

$$f(tx_1 + (1-t)x_2) \leq tf(x_1) + (1-t)f(x_2) \text{ for all } x_1, x_2 \text{ and } t \in [0,1].$$



- Convex function has a unique minimum.
- The local minimum is the global minimum.
- Convex function: $(x-1/2)^2$, e^x, $|x|$.
- Nonconvex function: $-(x-1/2)^2$, $\sin(x)$.

Subgradient

- u is said to be a subgradient of a function f at point x if

$$f(y) - f(x) \geq u_x^\top (y - x), \text{ for all } y.$$

- If f is differentiable, then $u_x = \nabla f(x)$ uniquely (It is a column.).
- If f is not differentiable, there could be many subgradients.
- $f(x) = |x|$, $u_x = \begin{cases} 1 & \text{if } x > 0; \\ -1 & \text{if } x < 0; \\ [-1, 1] & \text{if } x = 0. \end{cases}$
- **Proof:** Let Ω be the collection of subgradients of $f(x) = |x|$ at 0.
- If $u \in [-1, 1]$, we have $f(y) - f(0) = |y| \geq uy$ for all y.
- This shows $[-1, 1] \in \Omega$.
- If $u \in \Omega$, then by definition of subgradient and let $y = u$, we have

$$|u| = |y| = f(y) - f(0) \geq uy = u^2,$$

  which implies $|u| \leq 1$. This proves $\Omega \subset [-1, 1]$.

Subgradient

- $f(x) = \|x\|_2 = \sqrt{x^\top x}$, $u_x = \begin{cases} \frac{x}{\|x\|_2} & \text{if } x \neq 0; \\ \{z : \|z\|_2 \leq 1\} & \text{if } x = 0. \end{cases}$
- $f(x) = \|x\|_1 = \sum_{i=1}^p |x_i|$,

$$u_{x,i} = \begin{cases} 1 & \text{if } x_i > 0; \\ -1 & \text{if } x_i < 0; \\ [-1, 1] & \text{if } x_i = 0, \end{cases} \quad \text{and} \quad u_x = \begin{pmatrix} u_{x,1} \\ \vdots \\ u_{x,p} \end{pmatrix}.$$

- For simplicity, we still denote subgradient as $\nabla f(x)$ (It could be a set.).

Unconstrained Optimization

- Consider the optimization problem: $\min_{x \in \mathbb{R}^p} f(x)$.
- A saddle point $\widetilde{x}$ satisfy $0 \in \nabla f(\widetilde{x})$.
- $f(x) = \sin(x)$, $\nabla f(x) = \cos(x)$.
- Saddle points are $\{\ldots, -\frac{3\pi}{2}, -\frac{\pi}{2}, \frac{\pi}{2}, \frac{3\pi}{2}, \ldots\}$
- Global minimizers $x^* = \operatorname{argmin}_{x \in \mathbb{R}^p} f(x)$ are among the saddle points.
- $x^* = \{\ldots, -\frac{\pi}{2}, \frac{3\pi}{2}, \ldots\}$, multiple global minimizers.
- When f is convex, there is one saddle point, which is the global minimizer.
- $f(x) = \|x\|_1 = \sum_{i=1}^p |x_i|$,

$$u_{x,i} = \begin{cases} 1 & \text{if } x_i > 0; \\ -1 & \text{if } x_i < 0; \\ [-1, 1] & \text{if } x_i = 0, \end{cases} \quad \text{and} \quad \nabla f(x) = \begin{pmatrix} u_{x,1} \\ \vdots \\ u_{x,p} \end{pmatrix}.$$

- $0 \in \nabla f(x)$ implies $0 \in u_{x,i}$ for all $i = 1, \ldots, p$.
- Saddle points are $\{0\}$, which is also the global minimal.

Constrained Optimization

- Consider the optimization problem:

$$\begin{aligned} & \min_{x \in \mathbb{R}^p} f(x) \\ \text{subject to} \quad & h_i(x) \le 0, \text{ for } i = 1, \ldots, m, \\ & g_j(x) = 0, \text{ for } j = 1, \ldots, r. \end{aligned}$$

- Lagrange Multiplier: $\min_{x \in \mathbb{R}^p} f(x) + \sum_{i=1}^m u_i h_i(x) + \sum_{j=1}^r v_j g_j(x)$.
- Karush-Kuhn-Tucker (KKT) Conditions:
  Saddle points (or KKT points) $(x, u_i's, v_i's)$ satisfy
  - $0 \in \nabla f(x) + \sum_{i=1}^m u_i \nabla h_i(x) + \sum_{j=1}^r v_j \nabla g_j(x)$;
  - $u_i h_i(x) = 0$ for all i;
  - $h_i(x) \le 0, g_j(x) = 0$ for all i, j;
  - $u_i \ge 0$ for all i.
- If f, h_i's are convex, and $g_j(x) = a_j^\top x - b_j$ is linear, we say this is a convex problem.
  - Convex problem has one KKT point.
  - Nonconvex problem could have multiple KKT points.
  - Global minimizer is among the saddle points.

Exercise

- $\min_{x \in \mathbb{R}} -(x - 1/2)^2$ subject to $x \in [-1, 1]$.
- $f(x) = -(x - 1/2)^2$ and $h(x) = |x| - 1$ (no g).
- KKT Conditions:
  - $0 \in -2x + 1 + u\nabla h(x)$;
  - $u(|x| - 1) = 0$;
  - $|x| - 1 \leq 0$;
  - $u \geq 0$.
- Recall $\nabla h(x) = \begin{cases} 1 & \text{if } x > 0; \\ -1 & \text{if } x < 0; \\ [-1, 1] & \text{if } x = 0. \end{cases}$ We consider three cases.

Exercise

- Case 1: $x > 0$.
- KKT Conditions:
  - $0 \in -2x + 1 + u$;
  - $u(x-1) = 0$;
  - $x - 1 \leq 0$;
  - $u \geq 0$.
- $u = 0$ implies $x = 1/2$.
- $x = 1$ implies $u = 1$.
- Saddle Points: $\begin{cases} x = 1/2; \\ u = 0; \end{cases}$, $\begin{cases} x = 1; \\ u = 1. \end{cases}$

Exercise

- Case 2: $x < 0$.
- KKT Conditions:
  - $0 \in -2x + 1 - u$;
  - $u(-x - 1) = 0$;
  - $-x - 1 \leq 0$;
  - $u \geq 0$.
- $u = 0$ implies $x = 1/2$.
- $x = -1$ implies $u = 3$.
- Saddle Points: $\begin{cases} x = -1; \\ u = 3. \end{cases}$

Exercise

- Case 3: $x = 0$.
- KKT Conditions:
  - $0 \in -2x + 1 + u[-1, 1]$;
  - $u(-1) = 0$;
  - $-1 \leq 0$;
  - $u \geq 0$.
- $u = 0$ implies $x = 1/2$.
- No saddle points.

Exercise

- The saddle points (x, u) are $(1/2, 0), (1, 1), (-1, 3)$.
- Check $\min_{x \in \mathbb{R}} -(x - 1/2)^2$ at $x = 1/2, 1, -1$.
- Global minimizer $x^* = -1$.
- Multiple saddle points, because $f(x) = -(x - 1/2)^2$ is not convex.

Exercise

- Convex problem

$$\begin{aligned} &\min_{x \in \mathbb{R}^n} -\sum_{i=1}^{n} \log(\alpha_i + x_i) \\ \text{subject to} \quad & x_i \geq 0, \text{ for } i = 1, \ldots, n, \\ & \sum_{i=1}^{n} x_i = 1. \end{aligned}$$

- $f(x) = -\sum_{k=1}^{n} \log(\alpha_k + x_k)$, $h_i(x) = -x_i$, $g(x) = \sum_{i=1}^{n} x_i - 1$
- KKT Conditions:
  - $0 \in -\frac{1}{\alpha_i + x_i} - u_i + v$ for all i;
  - $u_i(-x_i) = 0$ for all i;
  - $-x_i \leq 0$ for all i and $\sum_{i=1}^{n} x_i - 1 = 0$;
  - $u_i \geq 0$ for all i.

Exercise

- KKT Conditions:
  - $0 \in -\frac{1}{\alpha_i + x_i} - u_i + v$ for all i;
  - $u_i x_i = 0$ for all i;
  - $x_i \geq 0$ for all i and $\sum_{i=1}^n x_i = 1$;
  - $u_i \geq 0$ for all i.
- Case 1: $u_i = 0$
- $0 = -\frac{1}{\alpha_i + x_i} - u_i + v$ implies $x_i = \frac{1}{v} - \alpha_i$;
- We need $x_i \geq 0$, so $v \leq \frac{1}{\alpha_i}$.
- Case 2: $u_i > 0$
- $x_i = 0$ and $0 \in -\frac{1}{\alpha_i + x_i} - u_i + v$ imply $u_i = v - \frac{1}{\alpha_i}$.
- We need $u_i > 0$, so $v > \frac{1}{\alpha_i}$.
- $x_i = \begin{cases} \frac{1}{v} - \alpha_i & \text{if } v \leq \frac{1}{\alpha_i}; \\ 0 & \text{if } v > \frac{1}{\alpha_i}; \end{cases} = \max(0, \frac{1}{v} - \alpha_i)$.

Exercise

- Using $\sum_{i=1}^n x_i = 1$, we need $\sum_{i=1}^n \max(0, \frac{1}{v} - \alpha_i) = 1$.
- Change of variable $y = \frac{1}{v}$
- $\sum_{i=1}^n \max(0, y - \alpha_i) = 1$ is a piecewise linear function of y.
- For example: assuming $\alpha_1 \leq \alpha_2 \leq \ldots \leq \alpha_n$.
- It is a piecewise linear function in $[0, \alpha_1], [\alpha_1, \alpha_2], \ldots, [\alpha_{n-1}, \alpha_n], [\alpha_n, \infty)$.
- You can find the solution y^* by checking each interval.
- Next calculate $v^* = \frac{1}{y^*}$.
- So $x_i^* = \max(0, \frac{1}{v^*} - \alpha_i) = \max(0, y^* - \alpha_i)$.
- Example: $n = 3, \alpha_1 = 1, \alpha_2 = 2, \alpha_3 = 3$
  - $\max(0, y - 1) + \max(0, y - 2) + \max(0, y - 3) = 1$ gives $y^* = 2$, $v^* = 0.5$
  - $x_1^* = 1$, $x_2^* = 0$, $x_3^* = 0$.
  - Optimal value is $-\log(1 + 1) - \log(2) - \log(3) = -\log(12) = -2.484907$

CVXR/CVXPY

- CVX is a package for solving convex optimization problem.

```
rm(list=ls())
library(CVXR)
n=3
x=Variable(n)
alpha=c(1,2,3)
obj=-1*sum(log(alpha+x))
constr=list(x>=0,sum(x)==1)
prob=Problem(Minimize(obj), constr)
result=solve(prob)
```

```
> result$value
[1] -2.484907
> result$getValue(x)
             [,1]
[1,] 9.998993e-01
[2,] 1.006791e-04
[3,] 3.483607e-08
> result$getDualValue(constr[[1]])
             [,1]
[1,] 6.771686e-09
[2,] 6.906044e-05
[3,] 1.667105e-01
> result$getDualValue(constr[[2]])
[1] 0.5000438
```

Primal and Dual

- Primal problem:

$$
\begin{aligned}
& p^* = \min_{x \in \mathbb{R}^p} f(x) \\
\text{subject to} \quad & h_i(x) \le 0, \text{ for } i = 1, \ldots, m, \\
& g_j(x) = 0, \text{ for } j = 1, \ldots, r.
\end{aligned}
$$

- Lagrange Multiplier:

$$
L(x, u, v) = f(x) + \sum_{i=1}^{m} u_i h_i(x) + \sum_{j=1}^{r} v_j g_j(x),
$$

where $u = (u_1, \ldots, u_m)^\top, v = (v_1, \ldots, v_r)^\top$.

- Fact: $p^* = \min_x \max_{u \ge 0, v} L(x, u, v)$

Primal and Dual

- Fact: $p^* = \min_x \max_{u \geq 0, v} L(x, u, v)$
- Proof: Case 1: If x is feasible, then $h_i(x) \leq 0$ and $g_j(x) = 0$ for all i, j. We have $\max_{u \geq 0, v} L(x, u, v) = f(x)$.
- Case 2: If x is not feasible, there is a i such that $h_i(x) > 0$ or a j such that $g_j(x) \neq 0$. For simplicity, we assume that $h_1(x) > 0$. We have

$$\max_{u \geq 0, v} L(x, u, v) \geq f(x) + \max_{u_1 \geq 0} u_1 h_1(x),$$

  which can be arbitrarily large. Hence, We have $\max_{u \geq 0, v} L(x, u, v) = \infty$.
- Combining above, we have

$$\max_{u \geq 0, v} L(x, u, v) = \begin{cases} f(x) & \text{if } h_i(x) \leq 0, g_j(x) = 0 \text{ for all } i, j, \\ \infty & \text{else.} \end{cases}$$

- Taking minimum over x, we complete the proof.

Primal and Dual

- Primal problem: $p^* = \min_x \max_{u \geq 0, v} L(x, u, v)$
- Define: $\phi(u, v) = \min_x L(x, u, v)$
- Dual problem: $d^* = \max_{u \geq 0, v} \phi(u, v) = \max_{u \geq 0, v} \min_x L(x, u, v)$
- Weak duality: $d^* \leq p^*$
- Proof: By definition, we have

$$\min_x L(x, u, v) \leq L(x, u, v)$$

$$\max_{u \geq 0, v} \min_x L(x, u, v) \leq \max_{u \geq 0, v} L(x, u, v)$$

$$\max_{u \geq 0, v} \min_x L(x, u, v) \leq \min_x \max_{u \geq 0, v} L(x, u, v)$$

- maxmin is less than minmax
- Weak duality always holds for convex or nonconvex problems.

Exercise

- Primal problem: $p^* = \min_x -(x-1/2)^2$ subject to $x \in [-1, 1]$.
- From previous slides, we have $p^* = -9/4$ when $x = -1$.
- Reformulation: $p^* = \min_x -(x-1/2)^2$ subject to $x - 1 \leq 0, -x - 1 \leq 0$.
- $L(x, u) = -(x-1/2)^2 + u_1(x-1) + u_2(-1-x)$ and

$$\phi(u) = \min_x \left\{-(x-1/2)^2 + u_1(x-1) + u_2(-1-x)\right\} = -\infty.$$

- Dual problem: $d^* = \max_{u \geq 0} \phi(u) = -\infty$.
- So, we have $d^* < p^*$ due to nonconvexity of the primal problem.

Exercise

- Primal problem: $p^* = \min_x (x - 1/2)^2$ subject to $x \in [-1, 1]$.
- It is easy to see that $p^* = 0$ when $x = 1/2$.
- Reformulation: $p^* = \min_x (x - 1/2)^2$ subject to $x - 1 \leq 0, -x - 1 \leq 0$.
- $L(x, u) = (x - 1/2)^2 + u_1(x - 1) + u_2(-1 - x)$ and $\phi(u) = \min_{x \in \mathbb{R}} L(x, u)$.
- Solving $0 = \frac{\partial}{\partial x} L(x, u) = 2(x - 1/2) + u_1 - u_2 = 2x - 1 + u_1 - u_2$ gives

$$x = x_u := \frac{u_2 + 1 - u_1}{2}.$$

- Hence, after some algebra, we have

$$\phi(u) = L(x_u, u) = -\frac{(u_2 - u_1)^2}{4} - \frac{u_1 + 3u_2}{2}$$

- Dual problem: $d^* = \max_u \phi(u)$ subject to $u_1, u_2 \geq 0$.
- So, we have $d^* = p^* = 0$ due to convexity of the primal problem.

Why Dual?

- Dual problem is always convex even if the primal problem is not.
- From dual to primal:
  - We wish to find the solution of the primal by solving the dual.
  - This requires the strong duality: $d^* = p^*$.
  - Strong duality holds for many problems: convex or nonconvex
  - Nonconvex: strong duality is verified case by case (hard!)
  - Convex: next slides
- If Strong duality holds, we have

$$\max_{u \geq 0, v} \underbrace{\min_x L(x, u, v)}_{\phi(u,v)} = \min_x \max_{u \geq 0, v} L(x, u, v).$$

  - Dual solution: $(u^*, v^*) = \operatorname{argmax}_{u \geq 0, v} \phi(u, v)$.
  - Primal solution: $x^* = \underbrace{\operatorname{argmin}_x L(x, u^*, v^*)}_{\text{unconstrained optimization}}$.

Exercise

- Primal problem: $p^* = \min_x (x - 1/2)^2$ subject to $x \in [-1, 1]$.
- Lagrange: $L(x, u) = (x - 1/2)^2 + u_1(x - 1) + u_2(-1 - x)$
- Dual problem: $d^* = \max_{u \geq 0} \phi(u)$ with

$$\phi(u) = -\frac{(u_2 - u_1)^2}{4} - \frac{u_1 + 3u_2}{2}$$

- Dual solution: $u^* = (0, 0)^\top$
- Primal solution:

$$x^* = \operatorname*{argmin}_x L(x, u^*) = \underbrace{\operatorname*{argmin}_x (x - 1/2)^2}_{\text{unconstrained optimization}} = 1/2.$$

Strong Duality of Convex Problem

- Primal problem:

$$\begin{aligned} & p^* = \min_{x \in \mathbb{R}^p} f(x) \\ \text{subject to} \quad & h_i(x) \le 0, \text{ for } i = 1, \ldots, m, \\ & a_j^\top x - b_j = 0, \text{ for } j = 1, \ldots, r. \end{aligned}$$

- Dual problem:

$$\begin{aligned} & d^* = \min_{u \in \mathbb{R}^m, v \in \mathbb{R}^r} \phi(u, v) \\ \text{subject to} \quad & u_i \ge 0, \text{ for } i = 1, \ldots, m. \end{aligned}$$

- Slater condition:
  - $h_i(x)$'s are convex.
  - There is a x such that $h_i(x) < 0$ for all i and $a_j^\top x = b_j$ for all j.
- Strong duality: Slater condition implies $p^* = d^*$ or equivalently

$$\max_{u \ge 0, v} \min_x L(x, u, v) = \min_x \max_{u \ge 0, v} L(x, u, v).$$

Quadratic Programming

- Equality-Constrained Quadratic Programming (EQP):

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \; & f(x) = \frac{1}{2} x^\top Q x + c^\top x \\ \text{subject to} \quad & Ax = b. \end{aligned}$$

- Here $Q \in \mathbb{R}^{n \times n}$ is symmetric, $c \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$.
- Let $A = \begin{pmatrix} a_1^\top \\ \vdots \\ a_m^\top \end{pmatrix}$ and $b = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}$, so $Ax = \begin{pmatrix} a_1^\top x \\ \vdots \\ a_m^\top x \end{pmatrix} = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} = b$

Quadratic Programming

- EQP can be written as

$$\begin{aligned} \min_{x \in \mathbb{R}^n} f(x) &= \frac{1}{2} x^\top Q x + c^\top x \\ \text{subject to} \quad & a_i^\top x = b_i \text{ for } i = 1, \ldots, m. \end{aligned}$$

- It is a convex problem.
- KKT Conditions:

$$\begin{aligned} 0 &= Qx + c + \sum_{i=1}^{m} \lambda_i a_i, \\ 0 &= a_i^\top x - b \text{ for } i = 1, \ldots, m. \end{aligned}$$

- Let $\lambda = (\lambda_1, \ldots, \lambda_m)^\top \in \mathbb{R}^m$, we can see

$$\begin{aligned} Qx + A^\top \lambda &= -c, \\ Ax - 0^\top \lambda &= b. \end{aligned}$$

Quadratic Programming

- The KKT point satisfies $\begin{pmatrix} Q & A^\top \\ A & 0 \end{pmatrix} \begin{pmatrix} x \\ \lambda \end{pmatrix} = \begin{pmatrix} -c \\ b \end{pmatrix}$.
- We can solve this problem using iterative algorithm.
- Example:

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \; & f(x_1, x_2) = x_1^2 + 2x_2^2 \\ \text{subject to} \quad & x_1 + x_2 = 5. \end{aligned}$$

- $f(x_1, x_2) = \frac{1}{2} x^\top \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix} x$ and $(1, 1)x = 5$.
- So $Q = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$, $c = 0$, $A = (1, 1)$ and $b = 5$.
- Solveing $\begin{pmatrix} 2 & 0 & 1 \\ 0 & 4 & 1 \\ 1 & 1 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \lambda \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 5 \end{pmatrix}$ leas to $x_1^* = \frac{10}{3}, x_2^* = \frac{5}{3}, \lambda^* = -\frac{20}{3}$.

Quadratic Programming

- General QP:

$$\begin{aligned} &\min_{x \in \mathbb{R}^n} f(x) = \frac{1}{2} x^\top Q x + c^\top x \\ \text{subject to} \quad &a_i^\top x = b_i \text{ for } i = 1, \ldots, m, \\ &a_j^\top x \leq b_j \text{ for } j = m+1, \ldots, m+p. \end{aligned}$$

- We say x is feasible if x satisfies the above constraints.
- Active set of x: $\mathcal{A}(x) = \{1, \ldots, m\} \cup \{j : a_j^\top x = b_j, j = m+1, \ldots, m+p\}$.

Active Set Algorithm (Optional)

- Idea: Choose an initial feasible $x_{(0)}$, let $W_0 = \mathcal{A}(x_{(0)})$ be its active set.
- We try to move $x_{(0)}$ to $x_{(0)} + d_{(0)}$ for some direction $d_{(0)}$.
- Finding $d_{(0)}$: Since $x_{(0)}$ is known, we can treat it as a constant, and consider

$$\begin{aligned} \min_{d \in \mathbb{R}^n} \; & f(d) = \frac{1}{2}(x_{(0)} + d)^\top Q (x_{(0)} + d) + c^\top (x_{(0)} + d) \\ \text{subject to} \quad & a_i^\top (x_{(0)} + d) = b_i \text{ for } i \in W_0. \end{aligned}$$

- Notice $f(d) = \frac{1}{2} d^\top Q d + (x_{(0)}^\top Q + c^\top) d + \text{ something free of } d.$
- Also $a_i^\top x_{(0)} = b_i$ as $i \in W_0$.
- The problem becomes

$$\begin{aligned} \min_{d \in \mathbb{R}^n} \; & f(d) = \frac{1}{2} d^\top Q d + (x_{(0)}^\top Q + c^\top) d \\ \text{subject to} \quad & a_i^\top d = 0 \text{ for } i \in W_0. \end{aligned}$$

- Let $d_{(0)}$ and λ^* be the solution of the above EQP.

Active Set Algorithm

- Moving: We consider several cases.
- Case 1: If $d_{(0)} \neq 0$, then
  - $a_i^\top (x_{(0)} + d_{(0)}) = b_i$ for all $i \in W_0$.
  - For some $i \notin W_0$, $x_{(0)} + d_{(0)}$ may violate the constraint $a_i^\top (x_{(0)} + d_{(0)}) \leq b_i$.
  - Try to move less: move $x_{(0)} + \gamma_{(0)} d_{(0)}$ for some $\gamma_{(0)} \in (0, 1]$.
  - How to choose $\gamma_{(0)}$?
  - For $i \notin W_0$, if $a_i^\top d_{(0)} \leq 0$, then you can choose any $\gamma_{(0)}$ because
  $$a_i^\top (x_{(0)} + \gamma_{(0)} d_{(0)}) = a_i^\top x_{(0)} + \gamma_{(0)} a_i^\top d_{(0)} \leq a_i^\top x_{(0)} \leq b_i.$$
  - For $i \notin W_0$, if $a_i^\top d_{(0)} > 0$, then we need find $\gamma_{(0)}$ satisfying
  $$a_i^\top (x_{(0)} + \gamma_{(0)} d_{(0)}) \leq b_i.$$
  - To do this, we need $\gamma_{(0)} \leq \frac{b_i - a_i^\top x_{(0)}}{a_i^\top d_{(0)}}$ for $i \notin W_0$ with $a_i^\top d_{(0)} > 0$.
  - Therefore, we choose
  $$\gamma_{(0)} = \min_{i \notin W_0, a_i^\top d_{(0)} > 0} \min(1, \frac{b_i - a_i^\top x_{(0)}}{a_i^\top d_{(0)}}).$$
  - Move $x_{(1)} = x_{(0)} + \gamma_{(0)} d_{(0)}$ and update $W_1 = \mathcal{A}(x_{(1)})$.

Active Set Algorithm

- Case 2: If $d_{(0)} = 0$, then we need to check the optimality.
  - Remember, the KKT conditions require the corresponding $\lambda_j^* \geq 0$ for inequality constraints.
  - If there are some $\lambda_j^* < 0$ for some inequality constraint, remove the index j with smallest λ_j^* from the Active set.
  - Move $x_{(1)} = x_{(0)}$ and update $W_1 = W_0 - \{j\}$.
- Repeat above procedures until $d_{(k)} = 0$ and the $\lambda_j^* \geq 0$ for all inequality constraints.

Active Set Algorithm

- Example:

$$\begin{aligned} &\min_{x\in\mathbb{R}^n} f(x_1,x_2) = x_1^2 + x_2^2 - 4x_1 - 4x_2. \\ \text{subject to} \quad & x_1 + x_2 \le 2, \\ & x_1 - 2x_2 \le 2, \\ & -x_1 - x_2 \le 1, \\ & -2x_1 + x_2 \le 2. \end{aligned}$$

- Matrix Form:

$$\begin{aligned} &\min_{x\in\mathbb{R}^n} f(x) = \frac{1}{2}x^\top \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} x + (-4,-4)x \\ \text{subject to} \quad & \begin{pmatrix} 1 & 1 \\ 1 & -2 \\ -1 & -1 \\ -2 & 1 \end{pmatrix} x \le \begin{pmatrix} 2 \\ 2 \\ 1 \\ 2 \end{pmatrix} \end{aligned}$$

- Choose $x_{(0)} = (0,-1)$, and $W_0 = \{2,3\}$.

Active Set Algorithm

- Original problem: $Q = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, c = \begin{pmatrix} -4 \\ -4 \end{pmatrix}, A = \begin{pmatrix} 1 & 1 \\ 1 & -2 \\ -1 & -1 \\ -2 & 1 \end{pmatrix}, b = \begin{pmatrix} 2 \\ 2 \\ 1 \\ 2 \end{pmatrix}.$
- Choose $x_{(0)} = (0, -1)$, and $W_0 = \{2, 3\}$.
- Solve

$$\begin{aligned} \min_{d \in \mathbb{R}^n} f(d) &= \frac{1}{2} d^\top Q d + (x_{(0)}^\top Q + c^\top) d \\ \text{subject to} \quad & a_i^\top d = 0 \text{ for } i \in \{2, 3\}. \end{aligned}$$

- The problem becomes

$$\begin{aligned} \min_{d \in \mathbb{R}^n} f(d) &= \frac{1}{2} d^\top \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} d + \left[(0, -1) \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} + (-4, -4) \right] d \\ \text{subject to} \quad & \begin{pmatrix} 1 & -2 \\ -1 & -1 \end{pmatrix} d = 0. \end{aligned}$$

Active Set Algorithm

- Simplify to standard EQP:

$$\begin{aligned} \min_{d \in \mathbb{R}^n} f(d) &= \frac{1}{2} d^\top \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} d + (-4, -6) d \\ \text{subject to} & \quad \begin{pmatrix} 1 & -2 \\ -1 & -1 \end{pmatrix} d = 0. \end{aligned}$$

- The solution:

$$\begin{pmatrix} d_{(0)} \\ \lambda \end{pmatrix} = \begin{pmatrix} 2 & 0 & 1 & -1 \\ 0 & 2 & -2 & -1 \\ 1 & -2 & 0 & 0 \\ -1 & -1 & 0 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 4 \\ 6 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ -0.667 \\ -4.667 \end{pmatrix}.$$

- Recall $W_0 = \{2, 3\}$, so $\lambda_2^* = -0.667$ and $\lambda_3^* = -4.667$.
- Remove 3 from the active set.
- $x_{(1)} = x_{(0)} = (0, -1)$ and update $W_1 = \{2\}$.

Active Set Algorithm

- Original problem: $Q = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, c = \begin{pmatrix} -4 \\ -4 \end{pmatrix}, A = \begin{pmatrix} 1 & 1 \\ 1 & -2 \\ -1 & -1 \\ -2 & 1 \end{pmatrix}, b = \begin{pmatrix} 2 \\ 2 \\ 1 \\ 2 \end{pmatrix}.$
- Choose $x_{(1)} = (0, -1)$, and $W_1 = \{2\}$.
- Solve

$$\begin{aligned} \min_{d \in \mathbb{R}^n} f(d) &= \frac{1}{2} d^\top Q d + (x_{(1)}^\top Q + c^\top) d \\ \text{subject to} \quad & a_i^\top d = 0 \text{ for } i \in \{2\}. \end{aligned}$$

- The problem becomes

$$\begin{aligned} \min_{d \in \mathbb{R}^n} f(d) &= \frac{1}{2} d^\top \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} d + \left[(0, -1) \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} + (-4, -4) \right] d \\ \text{subject to} \quad & \begin{pmatrix} 1 & -2 \end{pmatrix} d = 0. \end{aligned}$$

Active Set Algorithm

- Simplify to standard EQP:

$$\begin{aligned} \min_{d \in \mathbb{R}^n} f(d) &= \frac{1}{2} d^\top \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} d + (-4, -6) d \\ \text{subject to} & \begin{pmatrix} 1 & -2 \end{pmatrix} d = 0. \end{aligned}$$

- The solution:

$$\begin{pmatrix} d_{(1)} \\ \lambda \end{pmatrix} = \begin{pmatrix} 2 & 0 & 1 \\ 0 & 2 & -2 \\ 1 & -2 & 0 \end{pmatrix}^{-1} \begin{pmatrix} 4 \\ 6 \\ 0 \end{pmatrix} = \begin{pmatrix} 2.8 \\ 1.4 \\ -1.6 \end{pmatrix}.$$

Active Set Algorithm

- $d_{(1)} = (2.8, 1.4)^\top$ and $\gamma_{(1)} = \min_{i \notin W_1, a_i^\top d_{(1)} > 0} \min(1, \frac{b_i - a_i^\top x_{(1)}}{a_i^\top d_{(1)}})$.
- $x_{(1)} = (0, -1)^\top$ and $W_1 = \{2\}$.
- Original problem: $Q = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, c = \begin{pmatrix} -4 \\ -4 \end{pmatrix}, A = \begin{pmatrix} 1 & 1 \\ 1 & -2 \\ -1 & -1 \\ -2 & 1 \end{pmatrix}, b = \begin{pmatrix} 2 \\ 2 \\ 1 \\ 2 \end{pmatrix}$.
- $a_1^\top d_{(1)} = 4.2$, $a_3^\top d_{(1)} = -4.2$, $a_4^\top d_{(1)} = -4.2$
- $\gamma_{(1)} = \min(1, \frac{b_1 - a_1^\top x_{(1)}}{a_1^\top d_{(1)}}) = \frac{2+1}{4.2} = \min(1, \frac{5}{7}) = \frac{5}{7}$.
- Move $x_{(2)} = x_{(1)} + \gamma_{(1)} d_{(1)} = (2, 0)^\top$.
- We can verify $W_2 = \mathcal{A}(x_{(2)}) = \{1, 2\}$.
- Or a simpler method: $\gamma_{(1)}$ is blocked by $i = 1$, update $W_2 = \{1, 2\}$.

Active Set Algorithm

- $x_{(2)} = (2, 0)^\top, W_2 = \{1, 2\}$.
- Repeat to next step, we have $\begin{pmatrix} d_{(2)} \\ \lambda^* \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 1.333 \\ -1.333 \end{pmatrix}$
- $\lambda_1^* = 1.333$ and $\lambda_2^* = -1.333$.
- Remove 2 from the active set.
- $x_{(3)} = x_{(2)} = (2, 0)^\top$ and update $W_3 = \{1\}$.

Active Set Algorithm

- $x_{(3)} = (2, 0)^\top, W_3 = \{1\}$.
- Repeat to next step, we have $\begin{pmatrix} d_{(3)} \\ \lambda^* \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \\ 2 \end{pmatrix}$
- Original problem: $Q = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}, c = \begin{pmatrix} -4 \\ -4 \end{pmatrix}, A = \begin{pmatrix} 1 & 1 \\ 1 & -2 \\ -1 & -1 \\ -2 & 1 \end{pmatrix}, b = \begin{pmatrix} 2 \\ 2 \\ 1 \\ 2 \end{pmatrix}$.
- $\gamma_{(3)} = \min_{i \notin W_3, a_i^\top d_{(3)} > 0} \min(1, \frac{b_i - a_i^\top x_{(3)}}{a_i^\top d_{(3)}})$.
- $a_2^\top d_{(3)} = -3$, $a_3^\top d_{(3)} = 0$, $a_4^\top d_{(1)} = 3$
- $\gamma_{(3)} = \min(1, \frac{b_4 - a_4^\top x_{(3)}}{a_4^\top d_{(3)}}) = \min(1, \frac{2+4}{3}) = \min(1, 2) = 1$.
- Move $x_{(4)} = x_{(3)} + \gamma_{(3)} d_{(3)} = (1, 1)^\top$.
- It can be verify that $W_4 = \mathcal{A}(x_{(4)}) = \{1\}$.
- Or a simpler method: $\gamma_{(4)}$ is not blocked by any i, update $W_4 = W_3 = \{1\}$.

Active Set Algorithm

- $x_{(4)} = (1,1)^\top, W_3 = \{1\}$.
- Repeat to next step, we have $\begin{pmatrix} d_{(4)} \\ \lambda^* \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$
- $\lambda^* > 0$ and $d_{(4)} = 0$, the optimality conditions hold.
- So the solution is $x^* = (1,1)^\top$.

"quadprog" Package

```
library(quadprog)
Q=c(2,0,0,2)
Q=matrix(Q,ncol = 2, byrow = T)
c=c(-4,-4)
b=c(2,2,1,2)
A=c(1,1)
A=rbind(A, c(1,-2))
A=rbind(A, c(-1,-1))
A=rbind(A, c(-2,1))
```

```
> Q
     [,1] [,2]
[1,]    2    0
[2,]    0    2

> A
  [,1] [,2]
     1    1
     1   -2
    -1   -1
    -2    1

> c
[1] -4 -4
```

"quadprog" Package

```
solve.QP(Q,-c, -t(A), -b,meq=0)
```

```
$solution
[1] 1 1

$value
[1] -6

$unconstrained.solution
[1] 2 2

$iterations
[1] 2 0

$Lagrangian
[1] 2 0 0 0

$iact
[1] 1
```

Projection

- We say a set Ω is convex if

$$\lambda x + (1-\lambda)y \in \Omega \text{ for all } x, y \in \Omega \text{ and all } \lambda \in [0,1].$$

- Let Ω be a convex set and x be a point, which could be in or not in Ω.
- We say x^* is the projection of x onto Ω if

$$x^* \in \Omega \text{ and } \|x - x^*\| \leq \|x - y\| \text{ for all } y \in \Omega.$$

  Here $\|a\|^2 = a^\top a$ is the Euclidean norm.
- In other word, x^* is the closest point to x among all the points in Ω.
- The convexity of Ω guarantee the uniqueness of x^*.
- If $x \in \Omega$, then $x^* = x$.
- We write $x^* = Pr(x)$.

QP for Projection

- Consider convex set

$$\Omega = \{y : a_i^\top y = b_i \text{ for } i = 1, \ldots, m, a_j^\top y \leq b_j \text{ for } j = m+1, \ldots, m+p\}.$$

- The projection of x can be rewritten as

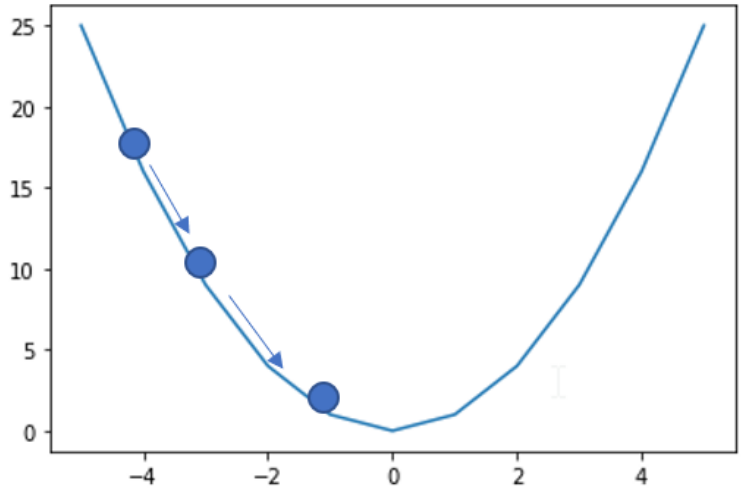
$$\begin{aligned} & \min_y f(y) = \|x - y\|^2 \\ \text{subject to} \quad & a_i^\top y = b_i \text{ for } i = 1, \ldots, m, \\ & a_j^\top y \leq b_j \text{ for } j = m+1, \ldots, m+p. \end{aligned}$$

- Notice $\|x - y\|^2 = y^\top y - 2x^\top y + x^\top x$.
- It is the equivalent to solve

$$\begin{aligned} & \min_y g(y) = \frac{1}{2} y^\top y - x^\top y \\ \text{subject to} \quad & a_i^\top y = b_i \text{ for } i = 1, \ldots, m, \\ & a_j^\top y \leq b_j \text{ for } j = m+1, \ldots, m+p. \end{aligned}$$

Gradient/Subgradient Descent

- Let f be a convex function and ∇f be one of its subgradient.
- Suppose x^* is the minimizer of f and $f^* = f(x^*)$ is the minimum.
- Algorithm: $x_{k+1} = x_k - \alpha_k \nabla f(x_k)$, here $\alpha_k > 0$ is the step size.
- Example:



- Choices of step size:
  - Constant step size: $\alpha_k = \alpha > 0$
  - Polyak's step size: $\sum_{k=0}^{\infty} \alpha_k = \infty$ and $\sum_{k=0}^{\infty} \alpha_k^2 < \infty$.
- Theoretically, x_0 can be chosen arbitrarily.
- The algorithm will stop when $|f(x_k) - f(x_{k+1})|$ or $\|\nabla f(x_k)\|$ is small.

Example

```
library(pracma)
f=function(x){
  s=1.2*(x-2)^2+3.2
  return(s)
}
gf=function(x){
  s=grad(f,x0=x)
  return(s)
}
gf=Vectorize(gf,"x")

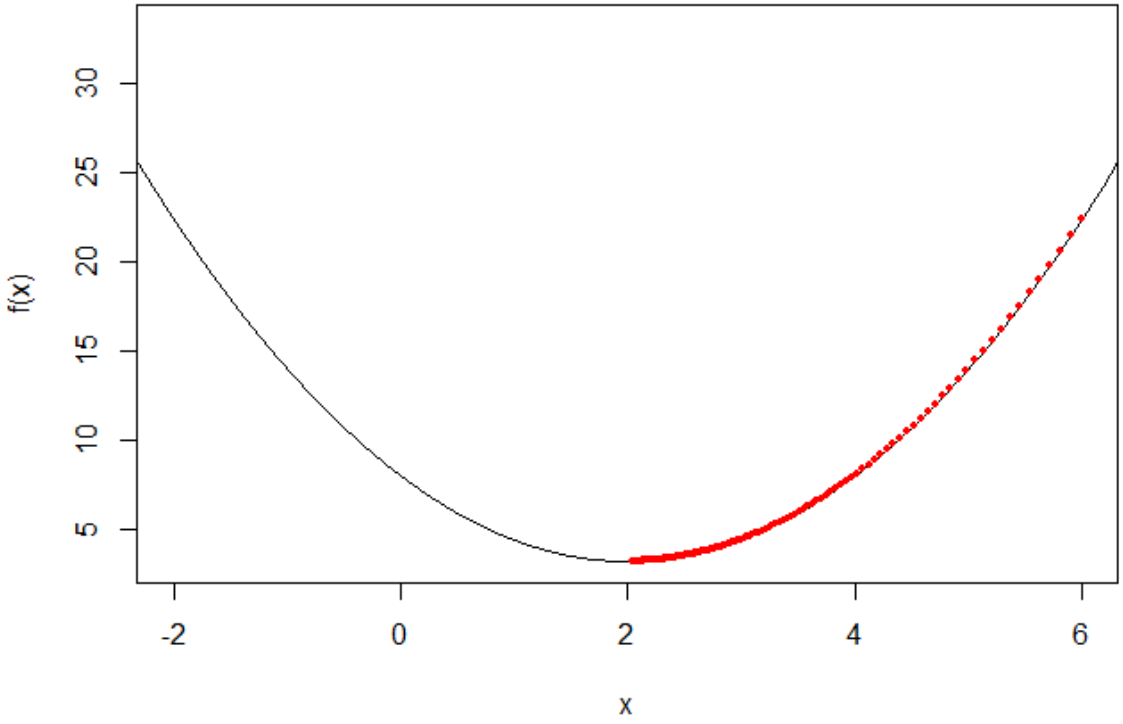
x=seq(-3,7,length.out=50)
plot(x,f(x),type="l",xlim=c(-2,6))

xk=6
alpha=0.01 #0.1, 0.6
error=1000
tol=10^(-4)
itmax=1000
it=1
while(it<=itmax & error>tol){
  points(xk, f(xk),pch=19,col="red",cex=0.5)
  xk_old=xk
  xk=xk-alpha*gf(xk)
  error=f(xk_old)-f(xk)
  error=abs(error)
  it=it+1
}
```

Example

- ```
  > xk
  [1] 2.040558
  ```

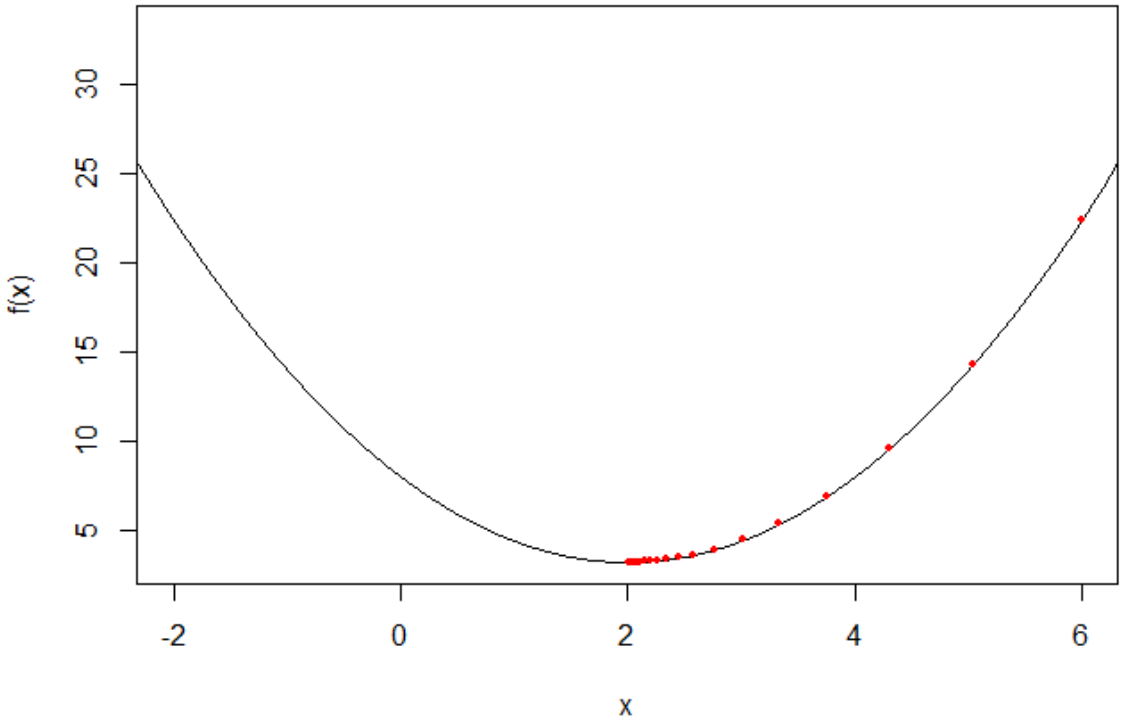
- Path with learning rate 0.01:


  ```

Example

- ```
  > xk
  [1] 2.009549
  ```

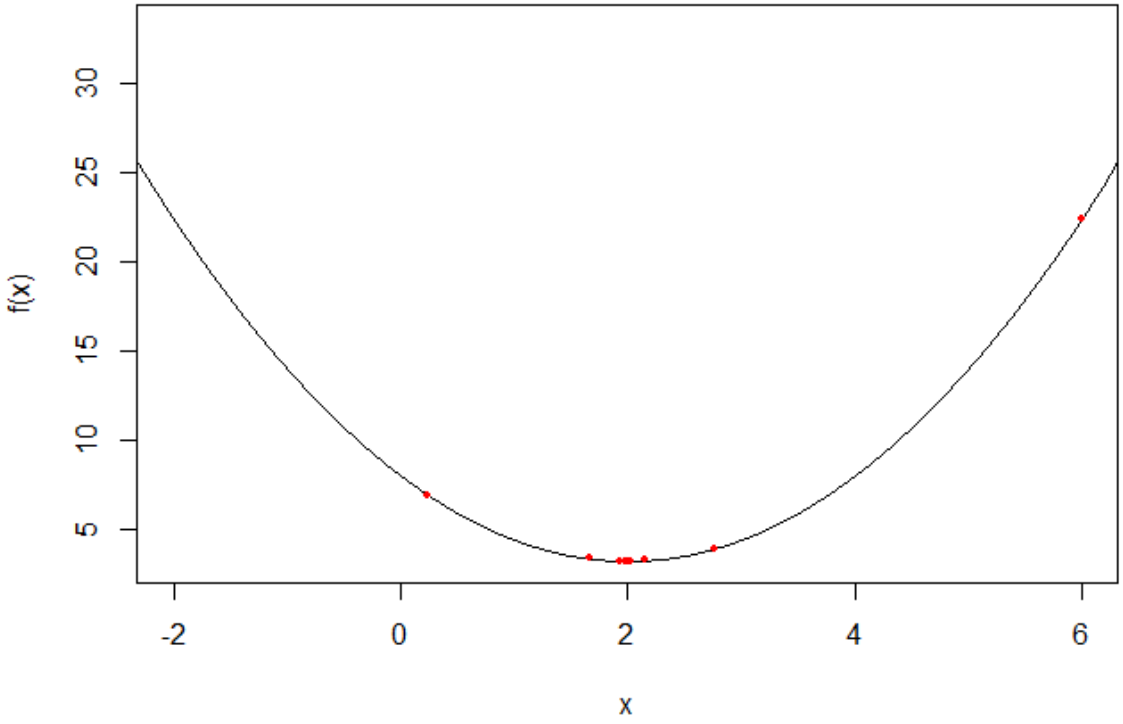
- Path with learning rate 0.1:


  ```

Example

- ```
  > xk
  [1] 1.997528
  ```

- Path with learning rate 0.6:


  ```

Projected Gradient/Subgradient Descent

- Let f be a convex function and ∇f be one of its subgradient.
- Let Ω be a convex set.
- Suppose x^* is the minimizer of f in Ω, that is

$$x^* = \operatorname*{argmin}_{x \in \Omega} f(x)$$

- Similarly, we define $f^* = f(x^*)$ to be the minimum.
- Algorithm: $x_{k+1} = Pr(x_k - \alpha_k \nabla f(x_k))$, here $\alpha_k > 0$ is the step size.
- The projection operator Pr forces x_k in Ω.

Optimization in statistical problems

- Statistical problem: $f(\beta) = \sum_{i=1}^{n} l(\beta, x_i, y_i)$, where (x_i, y_i) are observations.
- Linear regression: $f(\beta) = \sum_{i=1}^{n} (y_i - x_i^\top \beta)^2$.
- Logistic regression: $f(\beta) = \sum_{i=1}^{n} [\log(1 + e^{x_i^\top \beta}) - y_i x_i^\top \beta]$, with $y_i \in \{0, 1\}$.
- We can use gradient descent to solve the minimization problem.

Stochastic Gradient/Subgradient Descent

- If the objective function has the form $f(\beta) = \frac{1}{n}\sum_{i=1}^{n} l(\beta, x_i, y_i)$, we can use a stochastic algorithm.
- Stochastic Gradient Descent (SGD):
  - At the k-th iteration, randomly pick an index $i_k \in \{1, \ldots, n\}$
  - Updating $\beta_{k+1} = \beta_k - \alpha_k \nabla l(\beta_k, x_{i_k}, y_{i_k})$, here $\nabla l(\cdot, x_{i_k}, y_{i_k})$ is the subgradient of the function $\beta \to l(\beta, x_{i_k}, y_{i_k})$.
- Gradient Descent (GD): $\beta_{k+1} = \beta_k - \alpha_k \frac{1}{n}\sum_{i=1}^{n} \nabla l(\beta_k, x_i, y_i)$.
- GD uses all observations in each iteration, while SGD only uses one observation.
- In some literature, GD is called Batch Gradient Descent.
- Mini Batch Gradient Descent:
  - At the k-th iteration, randomly pick B indexes from $\{1, \ldots, n\}$ (with or without replacement).
  - Updating $\beta_{k+1} = \beta_k - \alpha_k \frac{1}{|B|}\sum_{s \in B} \nabla l(\beta_k, x_s, y_s)$,
- SGD is an example with $B = 1$, GD is the case with $B = n$.

Newton's method

- Goal: find $\operatorname{argmin} f(x)$ with f having second derivatives.
- Taylor expansion:

$$f(x) = f(a) + (x-a)^\top \nabla f(a) + \frac{1}{2}(x-a)^\top \nabla^2 f(a)(x-a) + \ldots$$

- Quadratic approximation:

$$f(x) \approx Q(x) = f(a) + (x-a)^\top \nabla f(a) + \frac{1}{2}(x-a)^\top \nabla^2 f(a)(x-a)$$

- Taking derivative

$$\nabla Q(x) = \nabla f(a) + \nabla^2 f(a)(x-a)$$

- $\nabla Q(x) = 0$ implies $x = a - (\nabla^2 f(a))^{-1} \nabla f(a)$
- Algorithm: $x_{k+1} = x_k - (\nabla^2 f(x_k))^{-1} \nabla f(x_k)$
- Gradient descent only uses first order condition of f, while Newton's method requires second differentiability.
- Unlike gradient descent, we do not need to rescale $f(x)$.

More Optimization Algorithms

- Adam (Adaptive Moment):
  - Most popular in pytorch
  - Use adaptively estimated first-order and second-order moments
- Proximal gradient method:
  - Achieve sparsity
  - Truth $(0, 1)$: sparse solution $(0, 0.9)$ V.S. nonsparse solution $(0.01, 1)$.
- DC (Difference of Convex) optimization:
  - $\min_x f(x) = f_1(x) - f_2(x)$, both $f_1(x), f_2(x)$ are convex.
  - $f(x)$ may not be convex.